Overview of Kalman Filter Theory with Applications to Navigation Wrap-Up

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Wrap-up

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- What have we learned?

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- Full-blown Kalman filtering is mathematically rich, employing concepts from caclulus, linear algebra, statistics, optimization

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- Kalman Filtering is really a kind of recursive averaging; recursive, weighted least squares, in fact
- Full-blown Kalman filtering is mathematically rich, employing concepts from caclulus, linear algebra, statistics, optimization
- Kalman Filtering is all about estimating the dynamic state of a system using all the information (or uncertainties) at our disposal

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- But there are workarounds for situations where the assumptions are violated
- Going beyond these assumptions is subject of much current research: Particle Filters, Unscented Kalman Filters, H_∞ Filters, higher order filters, machine learning techniques based on recurrent neural networks, general Bayesian filters, ensemble Kalman Filters (used in weather forecasting), smoothers, ...

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- Explore the Doxygen HTML files for the main OCX filter

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- Read the NAV ADD Appendix B re the Ensemble Timescale Filter developed by Dr. Ken Senior and Dr. Michael Coleman of NRL. Did you know that GPS Time is the product of a Kalman Filter?

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- Dynamic System; Process Model; State Transition Matrix, State Space and State Vector; Random Variable; Random Process; Process Noise; Covariance (P, Q, R); Measurement Model; White Noise; Random Walk; Gauss-Markov Process; Residual; SRIF

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- The Five Equations!!

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$$\textbf{Covariance Update: } P^{+}(k) = [I - K(k)H(k)]\,P^{-}(k) \tag{5}$$

$$\text{or } P^{+}(k) = [I - K(k)H(k)]\,P^{-}(k)\,[I - K(k)H(k)]^{\top} + K(k)R(k)K(k)^{\top}$$

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- Happy Trails! Until we meet again!